Artificial Intelligence and Machine Learning in Global Healthcare

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Abstract

Although medicine has been receptive to the benefits of machine learning and artificial intelligence (AI), it has only recently started to adopt this rapidly evolving, disruptive technology, particularly when compared to finance, entertainment, and transport sectors. Machine learning enables the detection of hidden connections and patterns, including outcomes prediction. Data is critical for the development of intelligent models, which harness the potential to improve and redefine disease self-management, treatment, and wellness pathways. The consequences of digital health democratization have significant health and ethical impact. This chapter provides an introduction to machine learning and AI and the development of intelligent healthcare systems. It explores applications of AI in healthcare and how the ubiquity of smartphones and Internet of Things (IoT) has accelerated the global shift from volume-based to value-based healthcare. This chapter will highlight key challenges within machine learning, evaluate machine learning projects, and share examples of best practice healthcare AI. Finally, it will review the ethical concerns surrounding machine learning, including how machines affect human behavior, data ownership, bias, unintended consequences, and the advances that have been made to support the global shift toward value-based population health.

Keywords

Digital health · Artificial intelligence · Big data · Value-based healthcare · Ethics

Introduction

There are more phones than people in the world, and the world is increasingly connected (ITU 2019). People use virtual assistants, self-driving cars, find partners through digital apps, and search the web for any symptom of ill-health. Each digital event leaves a digital exhaust that is "datafying" life as we know it. The success of many of the world's most-loved services, from Google to Uber, Alexa to Netflix, is grounded in big data, artificial intelligence (AI), and optimization.

Although medicine has been receptive to the benefits of big data, AI, and machine learning, it has been slow to adopt the rapidly evolving technology, particularly when compared to sectors such as finance, entertainment, and transport, that is, until recently (Alyass et al. 2015). Access to the amount of data necessary for useful healthcare AI and proven use cases only really emerged in the latter half of the last decade, subsequently blurring the line between data and AI.

The future of healthcare is in the midst of a significant transformation. The convergence of new digital big health data sources, smartphones, the ubiquity of 3G/4G/5G, inexpensive computing power, and the Internet of Things has catalyzed connected health solutions, data, and subsequently, real-world evidence. Whether connected through a wearable, health Internet of Things (IoT) device or app, people

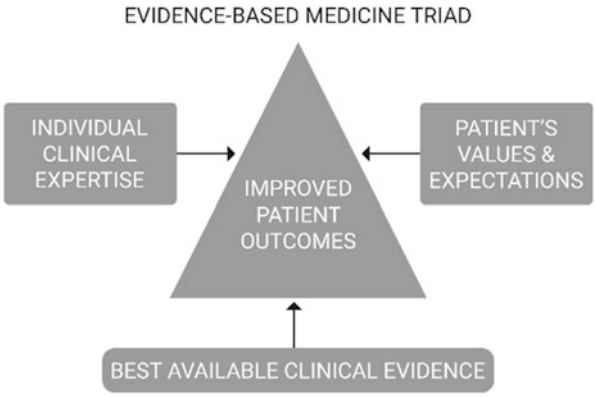


Fig. 1 Evidence-based medicine triad

can monitor their health in real-time. Healthcare is no longer provided just in the waiting room; it is truly mobile.

Healthcare strives to provide the best practice of evidence-based medicine. David Sackett, a pioneer of evidence-based medicine, defined the concept as "the conscientious explicit and judicious use of current best evidence in making decisions about the care of individual patients (Sackett 1997)." Evidence-based medicine is an approach to solve clinical problems by integrating best research and clinical evidence with real-world clinical expertise and patient values (Fig. 1).

Decision-making is a core tenet of evidence-based medicine. Decision-making should be based on the latest, most reliable scientific evidence. Data is vital in the making of these decisions.

Randomized controlled trials (RCTs) are considered the most robust and reliable form of evidence for assessing clinical efficacy. Evidence demonstrates that many factors can influence the reliability of RCTs, including methodological quality, reporting quality, and source of funding. Besides, studies have demonstrated bias in favor of industry in research funded by industry, which can undermine confidence in medical knowledge (Lesser et al. 2007).

Real-world evidence comes at a critical juncture for evidence-based medicine and is disrupting traditional evidence-based hierarchies and approaches. Datafication of human experience through smartphones, social media, digital communities, health apps, nutrition tracking, wearables, and health IoT has empowered patients to become their own evidence base and influence healthcare academia and understanding.

It is a necessity for evidence-based medicine to keep up with the developments of real-world evidence. As David Sackett said, "half of what you learn in medical school will be out of date in 5 years of your graduation; the trouble is nobody can tell you which half. The most important thing to learn is how to learn on your own" (Smith 2003).

Machine learning facilitates healthcare improvements using data, algorithms, and models to predict events and tailor interventions. Machine learning can support decision-making, personalize care, prioritize, and assist automation.

Health systems that leverage evidence-based medicine and machine learning models can engage patients and improve health outcomes. Across clinical, operational, and financial areas, machine learning can help staff with limited time target prioritize their efforts, give indications as to how a health system will perform in the future, anticipate outcomes, accelerate the identification of over- and underperforming subgroups, and automate routine classification tasks. Predictive tools can evaluate health risks and develop personalized healthcare plans to mitigate patient health risks, prevent disease, manage disease, and treat a disease precisely if it occurs (Cousins et al. 2002).

Data provides a wealth of opportunity for AI and machine learning. Exploration of this data over the ensuing decades will significantly progress healthcare and facilitate precision medicine.

At the same time, the world's population is living longer and is unhealthier than ever and in financial crisis. Healthcare services are returning to value-based and incentivized care, motivated by the surge in noncommunicable diseases such as type 2 diabetes, obesity, and cancer (Arnold et al. 2016; Yach et al. 2006).

Information is empowering, and data is proving an invaluable tool in improving health outcomes and value (D'Alessandro and Dosa 2001; Bernardi and Wu 2017). Data science, the science of big data, its analysis, and intelligent programming layers are now becoming pillars of healthcare systems.

However, data insight and real-world evidence are developing at a pace at which regulators are struggling. The consequences of digital health democratization have significant ethical implications. Big data, AI, and machine learning enable stakeholders to uncover hidden connections and patterns, including predictions of the future. The effects of understanding this data have moral and legal consequences that require appropriate governance to mitigate clinical risk and harm.

A High-Level Overview of AI Applications

AI is nothing new. At its core, AI is merely programming and an abstraction of computer science. AI's resurgence has much to do with the explosion of data from mobile devices, smartwatches, wearables, and the ability to access computer power cheaper than ever before. IBM estimated in 2011 that 90% of global data had been created in the preceding 2 years (IBM 2018).

AI can be defined as the simulation of intelligent behavior in agents (computers) in a manner that we, as humans, would consider being smart or humanlike. The core concepts of AI include agents developing traits including knowledge, reasoning, problem-solving, perception, learning, planning, and the ability to manipulate and move. There are four distinct categories of AI: reactive machines, limited memory, theory of mind, and self-aware AI.

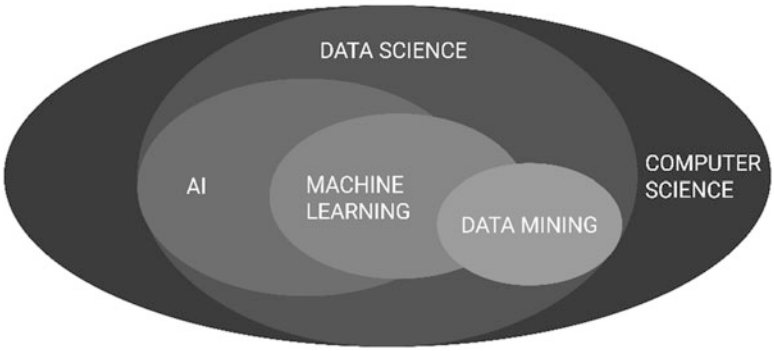


Fig. 2 AI, machine learning, and their place in computer science

AI can be considered to comprise the following:

- Getting a system to reason rationally. Techniques include automated reasoning, proof planning, constraint solving, and case-based reasoning.
- Getting a program to learn, discover, and predict. Techniques include machine learning, data mining (search), and scientific knowledge discovery.
- Getting a program to play games. Techniques include minimax search and alpha-beta pruning.
- Getting a program to communicate with humans. Techniques include natural language processing (NLP).
- Getting a program to exhibit signs of life. Techniques include genetic algorithms.
- Enabling machines to navigate intelligently in the world. This involves robotic techniques such as planning and vision.

In the past, the intelligence aspect of AI has been stunted due to limited datasets, representative samples of data, and the inability to both store and subsequently index and analyze considerable volumes of data. Today, data is available in real-time, from a variety of growing sources (Fig. 2).

Machine Learning

Systems that have hard-coded knowledge bases will typically experience difficulties in new environments. Difficulties can be overcome by a system that can acquire its own knowledge. This capability is known as machine learning and requires skills including knowledge acquisition, inference, updating and refining the knowledge base, and acquisition of heuristic – similar to human learning. Machine learning is an application of AI born from pattern recognition and the theory that computers can learn without being programmed to perform particular tasks.

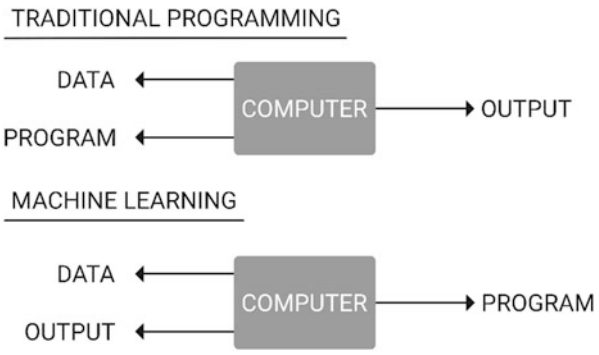


Fig. 3 Traditional programming compared to machine learning

Machine learning includes techniques such as Bayesian methods, neural networks, inductive logic programming, explanation-based, natural language processing, decision tree, and reinforcement learning.

Machine learning is a term credited to Arthur Samuel of IBM, who in 1959, proposed that it may be possible to teach computers to learn everything they need to know about the world and how to carry out tasks for themselves (Simon 1983).

Machine learning systems are not uncommon in healthcare. Researchers at Houston Methodist Research Institute in Texas developed software that reviews and translates mammograms 30 times faster than a human doctor and with 99% accuracy (Patel et al. 2017). According to the American Cancer Society, 50% of the 12.1 million mammograms performed annually yield false results meaning that healthy women get told they have cancer (American Cancer Society 2018). The software, which can be understood as an autonomous entity, learned from millions of mammogram records, reducing the risk of false positives.

Data drives machine learning. Intelligence is acquired through the ability to make effective decisions based on the nature of the learning signal or feedback. The utility of decisions is evaluated against a goal. Machine learning focuses on the development of algorithms that adapt to the presentation of new data and discovery.

Machine learning exemplifies principles of data mining but is also able to infer correlations and learn from them to apply to new algorithms. The goal is to mimic the ability to learn in a human, through experience, and achieving the assigned task without, or with minimal, external (i.e., human) assistance.

Just as with learning in humans, machine learning has many approaches. At its most basic, things are memorized. Second, agents learn from extracting information (through reading, listening, learning new things). And third, agents learn from example. For example, if (human) students were being taught square numbers and were shown a set of numbers Y = {1, 4, 9, 16, . . .} and taught that Y = n*n, they would typically understand the notion of square numbers without being explicitly taught every square number (Fig. 3).

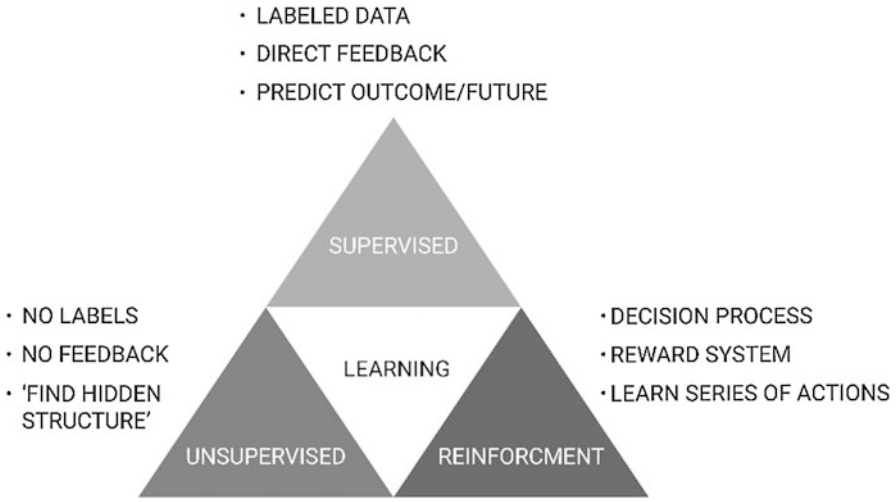


Fig. 4 Types of machine learning

Traditional software engineering and machine learning share a common objective of solving a problem or set of problems. The approach to problem-solving is what distinguishes the two paradigms.

Traditional software engineering or programming refers to the task of computerizing or automating a task such that a function or program, given an input, provides an output. In other words, writing a function f, such that given input x, output $y = f(x)$. This is done with logic, typically if-else statements, while loops and Boolean operators.

Machine learning differs from conventional programming in that rather than providing instructions about the function f, the computer is provided input x and output y and expected to determine or predict function f.

Whereas traditional programs have been written by humans to solve a problem, machine learning programs learn through reasoning to solve a problem from examples, rules, and information. Many classifiers in machine learning assist with prediction. Machine learning programs can also learn to generalize and help with issues of uncertainty due to the use of statistics and probability-driven techniques. Models can learn from previous computations or experience to produce reliable, repeatable decisions, and results.

Tasks for machine learning algorithms are categorized as (Fig. 4):

- Supervised learning (also known as inductive learning)
- Unsupervised learning
- Semi-supervised learning
- Reinforcement learning

In **supervised learning**, algorithms are presented with example inputs and the subsequently desired outputs in the form of training data, with the goal to learn general rules that map inputs to outputs. Input data is called training data and has a known output (result) associated with it. The training data guides the development of the algorithm. A model is created through a training process in which the model makes predictions and is corrected when predictions are incorrect.

Training continues until the model achieves the desired level of accuracy on the training data. Supervised algorithms include logistic regression. Supervised machine learning algorithms can apply what has been learned in the past to new data using labeled data examples to predict future events. The learning algorithm can also compare the output with the correct output to find any errors and modify the model accordingly.

Supervised learning problems can be employed in the following forms:

- **Classification**: To predict the outcome based on a training dataset where the output variable is in the form of distinct categories. Models are built through inputting training data in the form of prelabeled data. Classification techniques define decision boundaries and include support vector machines, naïve Bayes, Gaussian Bayes, k-nearest neighbors (KNN), and logistic regression. An example of classification is the label (or in real-world application, diagnosis) of someone as sick or unhealthy based on a set of symptoms.
- **Regression**: Very similar to classification. The only difference between classification and regression is that regression is the outcome of a given sample where the output variable is in the form of real values (for instance, blood oxygen SpO2 as a value rather than a classification of normal or abnormal). Linear regression, polynomial regression, support vector machine (SVR), ensembles, decision trees, and neural networks are examples of regression models.
- **Forecasting**: The method of making predictions based on past and present data. This is also known as time-series forecasting.
- An **ensemble** is a type of supervised learning that combines multiple, different, machine learning models to predict an outcome on a new sample. In an ensemble, models become features.

In **unsupervised learning**, the learning algorithm does not receive labels for data, leaving the algorithm to find structure from the input. As data is unlabeled, there is no evaluation of the accuracy of the structure that is output by the algorithm. Data may be missing both classifications and labels. As a result, the model is developed through interpretation: through finding hidden structures and inferences in the input data. When people reference systems that can learn for themselves, they are referencing unsupervised learning. Unsupervised learning techniques include clustering, dimensionality reduction, and association rule learning. The algorithm may never find the right output but instead models the underlying structure of the data. There are three types of unsupervised learning problems:

- **Association**: Discovering the probability of the co-occurrence of items in a collection. This is used extensively in marketing as well as healthcare.

For instance, this would be the percentage likelihood of developing any form of cancer if a smoker. Association is like classification; however, any attribute can be predicted in association problems, whereas classification is binary.

- **Clustering**: Grouping items such that items within the same cluster are more like each other than to items from another cluster.
- **Dimensionality reduction**: Dimensionality reduction can be achieved through feature selection and feature extraction. Dimensionality reduction may mathematically re-represent or transform data.

Feature extraction performs data transformation from a high-dimensional to a low-dimensional space. This can involve reducing the variables of a dataset while maintaining data integrity and ensuring that the most important information is represented.

Feature extraction methods can be used for dimensionality reduction. Typically, a new set of features would be created from the original feature set. An example would be combining all of a patient's clinical test results in a health risk scoring from health markers demonstrated to affect mortality. Reducing the number of dimensions allows easier visualization, particularly in two or three dimensions, as well as reducing time and space required for storage. Algorithms include the Apriori algorithm, FP-growth (also known as frequent pattern growth), Hidden Markov Model, principal component analysis (PCA), singular value decomposition (SVD), k-means, neural networks, and deep learning. Deep learning utilizes deep neural network architectures, which are types of machine learning algorithms.

Semi-supervised learning is a hybrid where there is a mixture of labeled and unlabeled data in the input. Although there may be a desired outcome, the model also learns structures to organize data and make predictions. Example problems are classification and regression. **Reinforcement learning** this is where systems interact with a dynamic environment in which an agent must perform a specific goal.

Reinforcement learning acts as an intersection between machine learning, behavioral psychology, ethics, and information theory. The algorithm receives feedback concerning rewards and punishments as it navigates the problem. Reinforcement learning allows the agent to decide the next best action based on its current state and by learning behaviors that will maximize the reward. Optimal actions (or an optimal policy) are learned through trial and error and feedback allowing the algorithm to determine ideal behaviors within context. Reinforcement learning is used in robotics, for instance, a robotic hoover that learns to avoid collisions by receiving negative feedback through bumping into tables, chairs, or other objects. Today, this approach also involves computer vision techniques. Reinforcement learning differs from standard supervised learning in that correct examples are never presented, nor are suboptimal decisions explicitly corrected. The focus is on performance in the real world (Fig. 5).

Like human learning, reinforcement learning assumes rational behavior regarding learning from one's experience and, as such, is assisting in understanding how humans learn from experience to make right decisions. DeepMind, for example, exposed their AlphaGo agent to master the ancient Chinese game of Go through

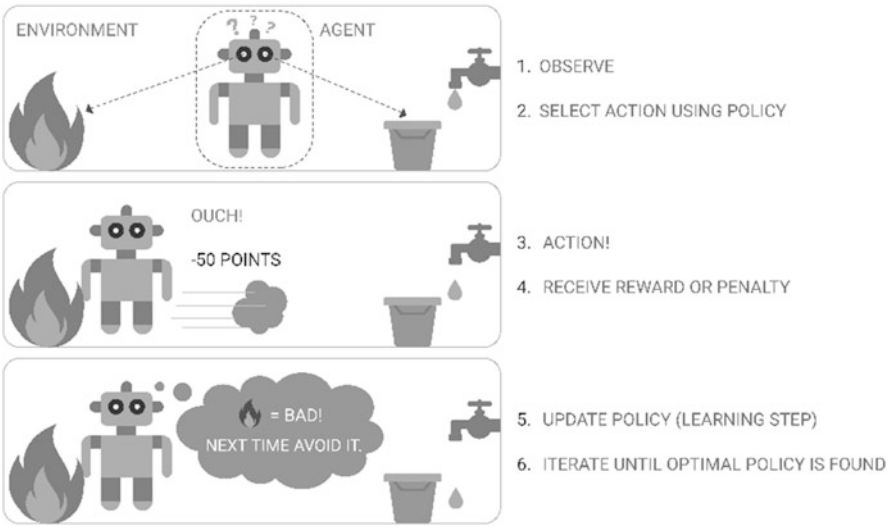


Fig. 5 Reinforcement learning example

showing "AlphaGo a large number of strong amateur games to help it develop its own understanding of what reasonable human play looks like. Then we had it play against different versions of itself thousands of times, each time learning from its mistakes and incrementally improving until it became immensely strong" (DeepMind 2019). Machines have overcome humans in the world of Go: world champion Go player Lee Sedol resigned from professional playing Go due to the "inevitability of AI systems" (Independent 2019b).

Reinforcement learning is used in autonomous vehicles. In the real world, an agent requires consideration of aspects of the environment, including current speed, road obstacles, surrounding traffic, road information, and operator controls. An agent learns to act in a way (or develop a policy) that is appropriate for the specific state of the environment. Governance of these actions leads to new ethical dilemmas posed by AI and machine learning for agents with reward-driven programming.

Data mining, also known as knowledge discovery in databases, is defined as the process of exploring data to extract novel information (including patterns and relationships) using sophisticated algorithms. Data mining can be applied to every aspect of healthcare and enables:

- Pattern predictions based on trends and behaviors
- Prediction based on probable outcomes
- Analysis of large datasets (particularly unstructured)
- Clustering through the identification of facts previously unknown

Data mining uses the power of machine learning, statistics, and database techniques and is at the heart of analytics efforts. Data mining typically uses large

amounts of data and can also give simpler, descriptive analytics. For instance, in healthcare, data mining applications can be used to identify conditions or medical procedures that are often seen together. Data mining also allows forecasting through identifying relationships within the data.

Applied in healthcare, machine learning, and AI are transforming the industry through improving diagnostics, predicting outcomes, and beginning to scratch the surface of personalized care. Examples of AI include:

- **Prediction**
  Technologies already exist that monitor data to predict disease outbreaks. This is often done using real-time data sources such as social media as well as historical information from the web and other sources. Malaria outbreaks have been predicted with artificial neural networks, analyzing data including rainfall, temperature, number of cases, and various other data points.
- **Diagnosis**
  Multiple organizations offer alternatives to non-emergency health systems, providing an AI doctor in the form of an AI chatbot. Combining the genome with machine learning algorithms provides the opportunity to learn about the risk of disease, improve pharmacogenetics, and provide better treatment pathways for patients.
- **Personalized treatment and behavior modification**
  A digital therapeutic, Low Carb Program, assists people with type 2 diabetes and prediabetes to self-manage the conditions and assist in placing them into remission. The app provides personalized education and integrated health tracking, learning from the user's and broader community's progress. At the end of a year, 54% of members who complete the app's education program reduce or eliminate medication dependency, achieving savings in medication deprescription (Saslow et al. 2018).
- **Drug discovery**
  The use of machine learning in preliminary drug discovery has the potential for various uses, from initial screening of drug compounds to predicted success rate based on biological factors. This includes R&D discovery technologies like next-generation sequencing. Drugs do not necessarily need to be pharmacological in their appearance. The use of digital solutions and aggregation of real-world patient data are providing solutions to conditions once considered to be chronic and progressive. Low Carb Program app, used by over 400,000 people with type 2 diabetes, places the condition into remission for 26% of the patients who complete the program at 1 year (Ibid.).
- **Follow-up care**
  Hospital readmittance is a considerable concern in healthcare. Doctors, as well as governments, are struggling to keep patients healthy, particularly when returning home following hospital treatment. Digital health coaches can provide a service similar to virtual customer service representatives on an e-commerce site. The assistant prompts questions about the patient's medications and reminds them to

take medicine, queries them about their condition symptoms, and conveys relevant information to the healthcare team.

Performing Machine Learning

Performing machine learning is similar to typical algorithm methodologies. It involves the following:

1. Specifying the problem as a learning task(s)
2. Preparing the data
3. Choosing the learning method
4. Applying the learning method
5. Assessing the method and results
6. Optimization
7. Reporting the results

The first step in performing machine learning is specifying the problem. Requirements for problem specifications, or scope, require understanding what the problem is, how the problem may be solved, and how to evaluate the problem's attempted answers (or models).

Machine learning algorithms learn from the data they are trained on. It is mission critical to provide the model with valid, correct data to learn. Data must be prepared in a usable format. In a real-world scenario, this would typically involve understanding the data that would be used to model the problem and exporting the data. The data must then be processed to ensure correct formatting, removal of erroneous data, and the fixing of any missing data. The dataset size may be more than required, and so dataset sampling may also be required. Every machine learning candidate will acquire skills in data preparation. Data pre-processing is essential to have tidy, valid data which is key to generate robust, veracious outcomes. This includes attribute selection, data transformations such as aggregation and decomposition, scaling, or weighting.

The algorithm chosen to develop the machine learning model depends on several factors, including the size, quality, and nature of data; the task deadline; and available resources and motivation for using the data. The learning technique is also referred to as the "representation of the solution," as each machine learning approach represents data differently. It is almost impossible to predict which approach will best perform on the data. Nevertheless, there are still general algorithms that apply to specific problems.

Machine learning tasks are typically conducted in a variety of programming languages: predominantly R, Python, MATLAB, and SQL. It is entirely possible to run a machine learning algorithm by implementing the ready-made algorithms provided by learning libraries such as scikit-learn, SciPy, Pandas, Matplotlib,

TensorFlow, and Keras. Google AI, Microsoft Azure, IBM Watson, and a host of other providers host platforms that can perform algorithms and the users to attempt a range of machine learning techniques on their uploaded data. One of the benefits of using such vendors is the ability to try more than one type of model for an experiment and compare the results, choosing the best solution for a problem task.

Select a test set and training set from the prepared data. A machine learning algorithm is trained on a "training dataset" and evaluated against the "test dataset." Some machine learning techniques may return one solution, whereas others may produce several. It is often a case of gathering their outputs (also referred to as hypotheses or learned models) and evaluating their outputs using methods such as predictive accuracy, comprehensibility, or utility. In most cases, Occam's razor is used, where the simplest solution is chosen if all else is equal.

- Predictive accuracy refers to the accuracy at which the agent performs the task of classification.
- Comprehensibility refers to how well we, as humans, can understand the output. For example, two Facebook bots recently began talking to each other in a language of their own (Independent 2018).
- Utility refers to the problem-specific measure of worth. For example, in drug synthesis, a combination of compounds that are unsafe for humans may be the best combination. However, this may not have the highest accuracy. Equally, this may override accuracy and comprehensibility.

After the assessment is complete, a candidate hypothesis is chosen, and report written.

The performance of machine learning tasks varies on the representation of the data given. For example, a patient record analyzed by an AI system does not examine the patient directly. Instead, data is fed into a system, with each piece of information relating to the representation of the patient known as a feature. It is not necessary to require complete feature sets as part of representations to have highly confident outputs. The goal of a machine learning algorithm is to generalize well, neither underfitting nor overfitting. Generalization refers to the model performing with maximum accuracy on instances not seen during training.

Generalization is measured by accuracy. Both under- and overfitting cause poor model performance and are detrimental to accuracy. If a model that was classifying with 92% accuracy in testing but only 30% on unseen data, we would say that the model does not generalize well from training to unseen data. This is overfitting. Overfitting is when a model fits the training data too well. It memorizes answers rather than learning concepts from them. In this case, the model cannot distinguish signal from noise and learns the noise present in the data. As a result, the model does not understand the actual patterns, or signals, in the data. As such, the model does not apply to new data and is determinantal to the model's ability to generalize. Decision trees are examples of algorithms that can overfit training data. Overfitting can be improved through pruning.

In many cases, it's a case of keeping things simple. Cross-validation is a technique to employ to measure against overfitting. Further still, training with more data may help algorithms infer hypotheses better.

The concept of "more data gives better results" is misguided. In many cases, more data may help. However, more data does not imply useful data. If data with noise were added, it would not help reduce overfitting. Hence, this is why it is paramount to ensure that data is clean and appropriate. Underfitting is simple to detect, as it will have poor performance. It refers to a model that cannot model the training data or generalize to new data. Typically, this would be represented by simple models that have too few features. Results would tend to show bias over variance (i.e., wrong outcomes).

Data

Data itself can take many forms – character, text, words, numbers, pictures, sound, or video. At its heart, data is a set of values of qualitative or quantitative variables. Each piece of data falls into two main types: structured and unstructured.

For data to become information, data requires interpretation. Information is organized or classified data, which has some meaningful value (or values) for the receiver. Information is the processed data on which decisions and actions should be based.

Structured data typically refers to something stored in a database, in a structure that follows a model or schema. Structured data is similar to machine language.

Within a clinical setting, EHRs may also take a similar, structured form. Readings from embedded sensors, smartphones, smartwatches, and IoT devices are typically forms of structured data – whether that be the provision of blood glucose readings, steps walked, calories burned, heart rate, or blood pressure.

Unstructured data refers to everything else. Unstructured data does not have a predefined model or schema. Data that is unstructured has no identifiable structure within it, and this presents problems for querying and information retrieval. E-mails, text messages, Facebook posts, Twitter tweets, and other social media posts are good examples of unstructured data (Fig. 6).

Gartner has forecasted that data volume is to grow 800% over the next 5 years, and 80% of this data will be in the form of unstructured data (Gartner 2018). A problem that unstructured data presents is one of volume as lack of structure makes compilation and interpretation a resource-intensive task consuming time and energy.

Structured data is traditionally more straightforward to analyze than unstructured data due to unstructured data's raw and unorganized form. However, analytics of unstructured, informal datasets is improving with the use of data science and machine learning methods such as natural language processing (NLP) which can assist in the understanding and classification of sentiment.

Semi-structured data does not necessarily conform with the formal structures of data schemas associated with relational databases or data tables.

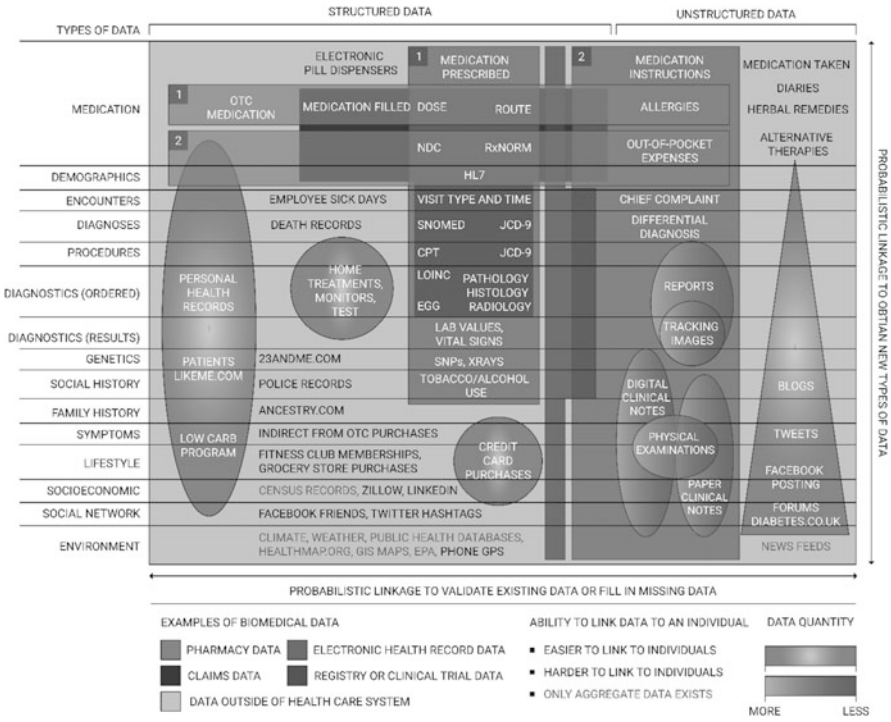


Fig. 6 Types of data (adapted from original source) (Weber et al. 2014)

However, semi-structured data may contain tags or markers to separate semantic elements and enforce grouping of records and fields within the data. Languages such as JSON (JavaScript Object Notation) and XML (Extensible Markup Language) are all forms of semi-structured data.

Data is the most critical component of successful AI. It is categorized into five sources:

- Web and social media data – clicks, history, health forums
- Machine-to-machine data – sensors, wearables
- Big transaction data – health claim data, billing data
- Biometric data – fingerprints, genetics, biomarkers
- Human-generated data – e-mail, paper documents, electronic medical records

The term big data is given to the collection of voluminous, traditional, and digital data that are sources for discovery and analysis. Big data is a popular term that defines datasets that are too big to be stored and processed in a conventional

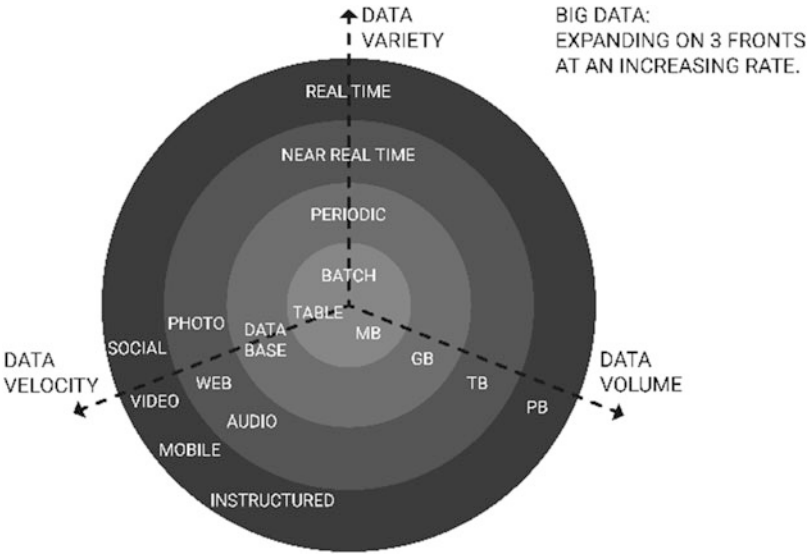


Fig. 7 Visualization of big data

relational database system. In this way, the term big data is vague – while the size is undoubtedly a part of big data, scale alone does not tell the whole story of what makes big data truly big.

The analytics of big data can uncover hidden patterns, unknown correlations, trends, preferences, and other information to make better and more informed decisions. Machine learning provides a toolbox of techniques that can be applied to datasets for this very purpose. Laney first described big data in 2001 as having the following characteristics (known as the 3 Vs) (Trifu and Ivan 2014):

- **Volume** – the quantity of generated and stored data. Big data is typically high in volume. The sheer size of data brings with it its intricacies in the form of storage, indexing, and retrieval.
- **Variety** – big data is varied in the types and nature of data, requiring efficient storage and analysis as well as systems for processing such data.
- **Velocity** – big data is received at a speed that brings its demands and challenges (Fig. 7).

Proximity is driving the future. As the scale of data increases, so too makes the demand that the time between data creation to insight and action be reduced. Just as data has evolved, so too has its analytics. The analysis of data has gone through several iterations, driven by the demand for access to data to drive decision-making.

Analytics 1.0 took the form of traditional analytics on small, slower velocity datasets, often in the form of spreadsheets and static documents. This is suitable for little data. Typical analytics were descriptive and report based.

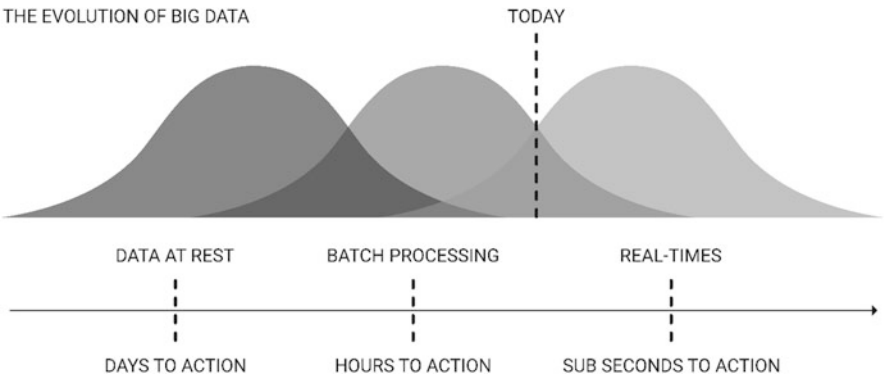


Fig. 8 Evolution of big data and its analytics timescale

Analytics 2.0 marked the integration of big data with traditional analytics, with interfaces to support the querying of big datasets in real-time. Big data analytics began in the 2000s, which facilitated competitive insight through complex queries combined with predictive views leveraging both structured and unstructured data such as social media, behavioral data, and the growing variety of user data. Big data that could not fit or be analyzed fast enough on a centralized platform would be processed with frameworks such as Hadoop, an open-source software framework for fast batch data processing across parallel servers both in the cloud and on-premise. Unstructured data would be supported through more than just SQL (Structured Query Language) databases, supporting key and value, document, graph, columnar, and geospatial data.

Analytics 3.0 marked the stage of maturity in which organizations realized measurable business impact from the combination of traditional analytics and big data. The approach supports agile insight discovery and real-time analysis at the point of decision. Analytics 3.0 is mostly a combination of conventional business intelligence, big data, and the IoT distributed throughout the network (Fig. 8).

Data's value is in the ability to convert it into information that can drive actionable insight to drive behaviors and workflows. Data analysis/analytics is a subset of data science that seeks to draw insights from sources of raw data. Today's advances in analyzing big data allow researchers to decode human DNA in minutes; predict where terrorists plan to attack; determine which gene is most likely to be responsible for specific diseases; and, of course, which ads users are most likely to respond to on Facebook.

Analysis of big data can help organizations understand the information contained in their data as well as identifying data most relevant to objectives, outcomes, and decisions.

Descriptive analytics brings insight to the past, focusing on the question of what happened, through analyzing data from history. Descriptive analytics uses techniques such as data aggregation and data mining to provide historical understanding. There is much to be learned from descriptive analytics.

Common examples of descriptive analytics are reports that provide the answers to questions such as how many patients were admitted to a hospital last July, how many were readmitted 30 days, and how many caught an infection or suffered from a mistake in care. Descriptive analytics serves as the first form of big data analysis.

Diagnostic analytics is a form of analytics that examines data to answer the question of why something happened. Diagnostic analytics comprises of techniques such as decision trees, data discovery, data mining, and correlations.

Predictive and prescriptive analytics requires the use of learning algorithms. Predictive analytics uses the data available and attempts to fill in missing data with best-guess estimates. Predictive analytics is characterized by techniques such as regression analysis, multivariate statistics, data mining, pattern matching, predictive modeling, and machine learning. Predictive analytics uses historical and current data to forecast the likelihood of future events or actionable outcomes.

Prescriptive analytics strives to make decisions for optimal outcomes: that is, to use all available data and analytics to inform and evolve a decision of what action to take – that is, smarter decisions. Prescriptive analytics attempt to quantify the effect of future decisions to advise on possible outcomes before decisions are made. At its best, prescriptive analytics predicts not only what will happen but also why it will happen to provide recommendations regarding actions that will take advantage of the predictions. Prescriptive analytics uses a combination of AI techniques and tools such as machine learning, data mining, and computational modeling procedures. These techniques are applied against input from many different datasets, including historical and transactional data, real-time data feeds, and big datasets. Prescriptive analytics performs an in-context prediction, considering the available evidence. For example, typical hospital readmission analytics display a forecast of patients likely to return in the next 14–30 days. A more useful predictor may integrate the same dashboard with projected costs, real-time bed counts, available education material, and follow-up care. With additional information, clinicians can determine patients with a high risk of readmission and take actions to minimize this risk, resulting in improved patient outcomes and reduced consumption of hospital resources.

Analysis of peoples' real-world health data can be analyzed at an individual level to create a patient data model and aggregated across populations to derive more considerable insights around disease prevalence, treatment, engagement, and outcomes. This improves the quality of care, transparency, outcomes, value, and at its core and democratizes healthcare delivery. By connecting real-world patient and clinical data to the genome, personalized medicine can be realized to the genetic makeup of patients and populations with application in the prescribing of medications, adverse effects and reactions, prevention strategies, and the likely risk of future diseases.

Although big data is defined by the fact it is stored in distributed systems, when it comes to creating machine learning algorithms, it is about quality and not quantity. Data-driven healthcare occurs when there is a synergetic partnership between healthcare providers, patients, and clinical and real-world data sources. This can

enable real-time analytics that drives actionable insights and allow for an enhanced patient experience and streamlined clinical and administrative workflows.

Volume-Based to Value-Based, Patient-Centered Healthcare

Healthcare itself is transitioning from volume- to patient-centric value (Sarwar et al. 2015). There is a shift in medicine in which pharmacology is not the first and only line of treatment, with a greater emphasis on the role of lifestyle as medicine for preventative and therapeutic purposes (Conrad 2015).

Traditional payment systems in healthcare are grounded in rewarding providers for the volume of patients referred to and treated. Volume-based care focuses on economies of scale. Providers receive a discrete amount of money for all of the services provided to patients for a set duration of time. Value for money, patient experience, and healthcare quality are subsequently secondary considerations for evaluation. Providers are incentivized by patient volume and the cost of the overall care provided to patients. Healthcare resource and cost pressures, exasperated doctors, and incentivized clinicians have unknowingly facilitated a "see as many patients as possible" approach that has driven disease management from a medication-first, patient-second perspective. Hospitals and clinicians are incentivized to see as many patients as possible, conduct as many tests as possible, and approach disease from a medication-first perspective. Volume-based care models are typically modeled on financial metrics such as optimizing profit or minimizing cost per patient rather than health outcomes.

Volume-based care payment systems often penalize providers financially for keeping people healthy, error reduction, or complication reduction, as they are not a driving key performance indicators (KPI). Personal biases can also complicate matters. In the UK, for example, doctors involved in assessing which drugs should be prescribed to NHS patients were found to have received money from pharmaceutical companies (Telegraph 2019). Volume-based care is focused on a finite amount of money for a population. With populations growing and resources stretched, the evolution toward value-based care has been expedited. Healthcare is now in a transition toward patient-based care, which is synonymous with value, rather than volume (Panesar 2019).

Tseng et al. defined the value in healthcare to be the quality of care, typically measured through healthcare outcomes (Jeffs et al. 2013). A value metric also for consideration is patient experience or patient-centeredness. Patient-centeredness is an essential but not necessarily dominant quality measure for value-based care. Patient-centered care utilizes a multifaceted approach, revolving around the patient, their goals, and the broader family in decisions and evaluations. There are several domains to patient-centered care, which includes patient-centeredness.

Patient-centered care is a collaborative and multifaceted approach involving delivering health and social care focused on patient experience and patient outcomes. Patient-centered care focuses on empowering patients with knowledge and experience, skills, and confidence to manage and optimize their health.

Care is compassionate, personalized, coordinated around the patient, and respectful of the patient's worldview. These goals may appear common sense for healthcare practitioners. However, they have not always been performed as standard practice. Healthcare has typically been reductionist in its approach, addressing a sum rather than the whole and acting for patients rather than with patients.

Two distinct philosophies, patient-centered care and value-based care, are becoming synonymous. Definitions of success incorporating patient-centered outcomes, perspectives, experiences, and preferences are positively influencing quality and delivery of care. Value-based care does not mean cheap or economy care. In fact, care is evaluated through the quality of patient experience and outcomes; and the model reshapes how patients receive care. Wellness and prevention are emphasized in value-based care. Lifestyle, behavior, and environmental factors account for 90% of disease risk (Anand et al. 2008). With growing pressure on healthcare resources, lifestyle medicine has silently refocused efforts on the role of lifestyle as medicine.

Evidence demonstrates that activity, nutrition, sleep, and stress all impact physical and emotional health markers (Stults-Kolehmainen and Sinha 2014). Lifestyle medicine's focus on lifestyle and behavior enables healthcare professionals to deal with reversible, noncommunicable diseases such as obesity and hypertension. Rather than treating the patient in a reductionist approach, the patient is provided with holistic, compassionate care. The prescription of lifestyle medicine is already taking place in healthcare, with digital applications prescribed to treat disease. Lifestyle interventions are typically safe, easy to implement, and scalable. Growing levels of chronic disease can be mitigated with a scalable approach delivered in an engaging and compelling setting.

Instead of reacting to events, there is a preventative, lifestyle-first approach to health in value-based care. The focus is on patient wellness, healthcare quality, and efficiency. This is where patient data, digital health, and AI hold a fountain of promise. The emphasis on efficiency ensures that the approach is scalable and effective, enabling providers to reduce healthcare costs and improve patient clinical health outcomes. Prevention of disease through quitting smoking, nutritional change, increased activity, better sleep, and identifying genetic risk factors reduces the burden on healthcare resources. Wellness is becoming incentivized, and it has become the best interest of insurance companies and healthcare providers to monitor their populations' health more closely. Start-ups and digital technology companies are developing digital health tools that are disrupting the traditional doctor-patient relationship and have enabled providers to become third actors involved in facilitating sustained health. The goal of value-based care is to standardize healthcare processes through best practices and to democratize access and quality of care. Mining of data and historical evidence can determine which methods work and which do not. Keeping people well reduces healthcare delivery costs and optimizes the use of resources.

For instance, when managing a chronic condition such as HIV, value-based care uses a collaborative and multidisciplinary approach to managing the disease and preventing complications. Patients engage with one healthcare team that is aware of patient progress and health status.

The healthcare team may contain the patient's nurse, dietician, behavior health coach, and other professionals to support patient progress.

Incentivization can be transformed. Take a hospital setting, for instance, in which rather than being paid on how many patients could be seen; the hospital is being compensated based on the opposite. The hospital is paid based on how many patients are in positive health status and how many beds are available. The focus moves from the point of hospital admission to predicting the likelihood of future risks and anticipating them with preventative solutions. As well as face-to-face delivery, ongoing patient support can be provided digitally, for example, with health coaching, delivering an exercise program, or helping with mental health concerns through apps, wearable technology, or telemedicine. AI and predictive analytics enable healthcare providers to optimize the delivery of healthcare, focusing on preventing and treating disease. Value-based care is utilizing low-cost digital technologies to enhance and democratize care. The simultaneous collection of behavioral, demographic, health, and engagement data provides an opportunity for machine learning and development of whole-system novel AI applications to rapidly improve, and learn from, user behavior and outcomes.

Managing healthcare operations and subsequently evaluating healthcare quality are complex. To assess a value-based care model, healthcare organizations must collect and analyze data, objectively assessing performance through the quality of care, patient health outcomes, and efficiency of cost. Providers can report and model preventative care metrics such as hospital readmission rates, error rates, disease progression, population health improvement, and engagement strategies. Several metrics define healthcare quality. Patient experience and satisfaction scoring are typically the first metric of evaluation. Metrics such as time spent per patient, patient engagement, medication savings, and adherence are examples of quality that relate to productivity.

Evaluating Machine Learning Projects

Typically, the most laborious tasks within a machine learning project are identifying the appropriate model and engineering features which both make a substantial difference to the output of the model. Features chosen can often have more impact on the quality of a model compared to the model choice itself. Therefore it is crucial to evaluate the learning algorithm that will determine the model's intelligence to predict the output of an unknown sample.

There are several stages of development and evaluation that take place to deploy a machine learning model, as illustrated successfully (Fig. 9).

The first stage is the prototype phase. During this phase, a prototype is created by testing various models on historical data to determine the best model. Hyperparameter tuning, as discussed later in this chapter, is a requirement of model training. Once the best prototype model is chosen, the model is tested and validated. Consider the fact that there is no such thing as a random dataset, and instead, the randomness applies to the splitting of the dataset. Be aware of biases that may appear

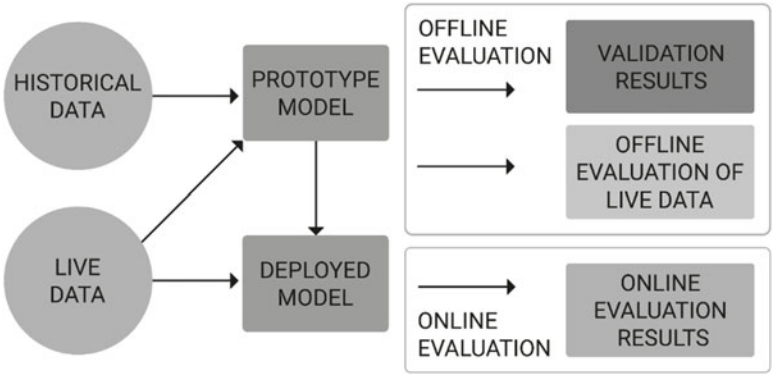


Fig. 9 Evaluation of machine learning models

in the data. Once the model has been successfully validated, it is deployed to production. The model is then usually evaluated by one (or several) performance metrics.

There are two ways of evaluating a machine learning model: offline evaluation and online (or live) evaluation.

A deployed machine learning model consumes data from two sources: historical data (or the data that is used as the experience to be learned from) and live data. Many machine learning models assume stationary distribution data – that the data distribution is constant over time. However, this is atypical of real life, as distributions of data often change over time – known as a distribution shift. For instance, consider a system that predicts the side effects of medications to patients based on their health profile. Medication side effects may change based on population factors such as ethnicity, disease profile, territory, medication popularity, and new medications. The distribution of relevant side effects based on patient data can vary quickly over time, and hence a model needs to detect a shift in distribution and accordingly evolve the model. The method in which this is typically assessed is through the performance of the model based on live data, evaluated through the validation metric used in the testing and validation of the model on historical data.

Model performance that is similar to or within a threshold of permissibility when evaluated on live data is deemed as a model that continues to fit the data. Degradation of model performance indicates that the model does not fit the data and requires retraining.

Offline evaluation measures the model based on metrics learned and evaluated from the historical, stationary, distributed dataset. Metrics such as accuracy and precision-recall are typically used within the offline training stage.

Online evaluation refers to the evaluation of metrics once the model is deployed. The key takeaway is that these metrics may differ from the metrics used to evaluate performance when the model is deployed live. For instance, a model that is learning on new pharmacological treatments may seek to be as precise as

possible in training and validation; but when placed online, it may need to consider business goals such as budget or treatment value when deployed. Online evaluation, particularly in the digital age, can support multivariate testing to understand best-performing models. Feedback loops are vital to ensuring that systems are performing as intended and help to understand the model in the context of use better. This can be performed by a human agent or automated through a contextually intelligent agent or user of the model.

It is essential that the evaluation of a machine learning model is based on a statistically independent dataset and not on the dataset it is trained on. This is because the evaluation of the training dataset is optimistic about the model's authentic performance as it adapts to the dataset. By evaluating the model with previously unseen data, there is a better estimate of the generalization error. New data can be hard to find; hence it is important to be able to have new, unseen data from the current dataset.

There are a plethora of evaluation metrics for a particular machine learning problem. Choosing the appropriate metric to evaluate a model depends on the use case – such as the impact of false positives, false negatives, and the consequences of such predictions. Furthermore, if a model is attempting to predict an event that only happens 0.001% of the time, an accuracy of 99.999% can be reported but not confirmed.

Multivariate testing is a handy method of determining which model is best for the particular problem at hand. Multivariate testing is known as statistical hypothesis testing and determines the difference between a null hypothesis and an alternate hypothesis. The null hypothesis is defined as the new model not affecting the average value of the performance metric, whereas the alternative hypothesis is that the new model does change the average value of the performance metric.

Although not a fail-safe, this reduces the change of illusionary results. If there is indeed a change between the null and alternate hypotheses, the difference will be confirmed.

Multivariate testing compares similar models to understand which is performing best or compares a new model against an older, legacy model. The particular performance metrics are compared, and a decision is made on which model to proceed with.

Case Studies: Examples of Healthcare AI

Low Carb Program: Digital Therapy for Type 2 Diabetes, Prediabetes, and Sugar Addiction

Low Carb Program is a behavior change platform available as a multi-platform app (iOS, Android, web, Alexa, Android Wear, Apple Watch, and VR) that provides goal-focused education and long-term behavior change support to patients with conditions related to hyperinsulinemia and metabolic syndrome. Approved for use by the NHS, the program has streams for type 2 diabetes, prediabetes, obesity, metabolic syndrome, polycystic ovarian syndrome (PCOS), nonalcoholic fatty liver disease (NAFLD), and general well-being. The user

experience (education, support, resources, interactions) is personalized to the user's health and demographic data and, as a result, provides a unique, individual experience to each user.

Peer-reviewed published 1-year outcomes demonstrate patients with type 2 diabetes who complete the program achieve a 7% weight loss, 13 mmol/mol A1c reduction, and 39% of patients place their A1c under the diagnostic threshold for type 2 diabetes (Saslow et al. 2018). Of type 2 diabetes patients starting the Low Carb Program on medication, 40% eliminated at least one medication from their treatment, 60% of patients starting on insulin reduced or eliminated it from their regime, and 26% of patients were classed as being in type 2 diabetes remission, that is, having an A1c under the diagnosis threshold without the use of medication.

As a digital therapeutic, Low Carb Program facilitates improved self-management of long-term health conditions through personalization of the learning and behavior change experience. What's more, as a digital therapy, the platform can assist users in achieving type 2 diabetes remission – something medication cannot. Through a data-driven approach and publication of big real-world evidence (data) to demonstrate real-world impact, the platform has redefined type 2 diabetes as not having to be a chronic and progressive disease at scale.

The impact on metabolic health is well evidenced. Nevertheless, through data collection, learning, and analysis, the impact of the platform on other aspects of human health is being evidenced.

Human research shows that sugar and sweetness can induce neurobiological reward circuits and craving that are comparable in magnitude to addictive drugs (DiNicolantonio et al. 2018), and food addiction has been implicated as a causal factor in chronic overeating, binge eating, and obesity (Corsica and Pelchat 2010). Food addiction also hampers weight loss, a critical factor in type 2 diabetes management (Elfhag and Rössner 2005). However, there is very little research in this area.

The prevalence of food addiction in a sample of type 2 diabetes patients joining the Low Carb Program was measured to understand if the provision of a digital, evidence-based, structured education and behavior change platform focused on real food, and a restricted sugar and carbohydrate plan could be a suitable intervention to manage food addiction. Users joining during April 2018 with a complete baseline dataset were asked to complete the Yale Food Addiction Scale. Users were followed for 6 months ($N = 283$; mean age 58.2, SD 17.7 years; 60% female; mean years since diagnosis 7.6, SD 6.7) and asked to complete the Yale Food Addiction Scale again at the 6-month follow-up point.

Of 283 study participants, 244 (86.2%) individuals reported outcomes at 6 months. At baseline, 76 (26.86%) met the diagnosis criteria for food addiction, as defined by the Yale Food Addiction Scale. An intention to treat analysis was conducted, showing a statistically significant decrease to 31 (10.95%) after 6 months of membership on the Low Carb Program ($p<0.0001$) suggesting participants who engage in the platform can effectively reduce symptoms of food addiction.

The Role of Online Health Communities in Patient Empowerment

The role of online communities in patients' empowerment and health self-management is little explored. A study by Royal Holloway, University of London, analyzed the impact of participating in the Diabetes.co.uk online community on members' well-being and how knowledge shared in the community influences patient-doctor interaction and the quality of healthcare patients receive. Data was collected through a survey of 1500 patients, a focus group, and 11 interviews. Findings demonstrated that the community could have a positive impact on its members' well-being and help members manage their relationship with healthcare professionals. Critical drivers of members' participation in the forum were understood, and the impact of the knowledge shared on the forum on members' sense of empowerment, their ability to manage diabetes, and the relationship with their doctor or nurse (Bernardi and Wu 2017). Findings from the focus group and interviews evidenced how the forum empowers members to take better care of their diabetes and discuss their needs with their doctor or nurse. Community forum members use knowledge from the forum to become empowered and provide further insights into the impact of the forum on members' relationship with their healthcare professionals.

Gro Health: Monitoring Patients with Hypertension in Real-Time with AI-Powered Physicians

Gro Health is a multi-platform modifiable risk app providing users with education, support, and behavior change activities to affect their activity/exercise, sleep, mental and physical well-being, and nutrition. Users are continuously engaged with the app through the curation of the user experience based on the user's registration health, demographic, and profile data. The platform is connected to a plethora of wearable and connected devices and has a real-time stream of patient big data from health IoT.

As a digital therapeutic, Gro provides a patient-facing and healthcare professional-facing ecosystem for the monitoring of patient health by healthcare professionals and AI in real-time for a variety of health conditions. Physicians prescribe Gro (i.e., refer the digital platform) to patients, and the patient's healthcare team subsequently monitors the health and engagement of the patent in real-time through a health analytics dashboard.

One such use case is with patients with hypertension. Patients are provided access to Gro and a wearable device enabling the patient's steps, heart rate, and heart rate variability data to be monitored. If a patient's data then subsequently follows a concerning trend (i.e., if the heart rate was to drop below/above-designated threshold), onboard AI algorithms send notifications to the healthcare professional team for investigation and triaging. In emergencies, the patient, the healthcare professional team, and emergency teams can be notified.

The consequence of an intelligent system such as this enables patients' health to be monitored in real-time by both AI and humans. At the same time, this poses a range of ethical AI and data privacy questions that require exploration.

Machine Learning Challenges

There are several key challenges in the development of machine learning initiatives.

Data Sources, Quality, and Growth

By the very nature of machine learning – data is required. Moreover, when it comes to big data, storage is a challenge. A significant challenge with big data is the volume and velocity at which data is generated and delivered. Managing the plethora of incoming data sources is a task in itself. There is broad agreement that the size of the digital universe will double every 2 years, which equates to a 50-fold growth from 2010 to 2020 (Lawlor 2017). Most big data is unstructured, meaning that it does not reside in a traditional database schema. Documents, photos, audio, videos, and other unstructured data can be awkward to search, analyze, and retrieve. Hence, the management of unstructured data is a growing challenge. Big data projects can evolve as quickly as the data used within them.

Data quality is critical. Common causes of dirty data must be addressed and include user input errors, duplicate data, and incorrect data linking. As well as maintaining and cleaning data, big data algorithms can also be used to help clean data.

Infrastructure

Big data consumes technical resources in the form of infrastructure, storage, bandwidth, and databases. The challenge is not technical so much as one of finding reliable vendors of services and support, and of getting the correct economic model of remuneration. A solution delivered from the cloud will have greater scalability, cost-effectiveness, and efficiency compared to an on-premises solution.

Governance

Data is a precious asset to any business or organization; in healthcare, data governance is paramount to ensure consistency, safety, and progression of any organization toward becoming data-driven and analytically driven. The need for data governance is driven primarily by the requirement for accountability in a risk-averse industry.

Governance of data refers to the practice of managing data assets throughout their life cycle to ensure that they meet standards of quality and integrity. The goal of data governance is to provide user trust in the data through optimal data validity and accountability. It sets the framework for creating high-quality data and using data in a secure, ethical, and consensual manner. Data governance in healthcare seeks to improve efficiency, create accountability, and establish a pool of data that

providers and patients can use to make proper health decisions. Often the best approach to data governance is to govern to the least extent necessary to achieve the most common good. In several cases, governance is being set for circumstances that do not require governing yet. Organizations can waste tremendous amounts of time through management and decision-making, including constraints on data, which can hinder project development and direction. Healthcare organizations should be motivated by the acquisition of more data to learn: to gather information to manage risks better and improve outcomes. Data governance is required to uphold a variety of criteria to return quality information.

The vast amount of data captured through wearable devices and sensors raises questions as to how data is collected, how data is treated, how data is analyzed, and how data should affect the policymaking process. Data privacy and security are protected areas in a state of constant development and evolution. Processes and standards such as ISO (International Organization for Standardization) and GDPR (General Data Protection Regulation) have been used to bring the entire data industry in line with best practice. Data governance and information governance are often used interchangeably but differ.

The **steward** is accountable for the data as defined and its appropriate use. The objective of this aspect of data governance is to assure quality regarding data accuracy, data accessibility, and data consistency, completeness, and updating.

Ensuring **data quality** is arguably the most critical function of data governance besides security. Poor data quality has a detrimental impact on accuracy, particularly for learning projects. Governance should seek to enforce quality, accuracy, and timeliness of data.

The **security of data** is essential. Data governance sets out how the data is protected, typically how it is encrypted, who has access to it, processes for handling data – including how the data may be used and what is done in case of a breach. Several drivers motivate data security: regulatory requirements that demand action (such as GDPR), the growing awareness of cybersecurity, user demand, and perceptions of digital weakness that can harm reputation and trust. The number of web sites that operate with a secure https:// protocol has risen by 4,000% since 2012, demonstrating acknowledgment for security (TechCrunch 2019). Patients expect their data to be secure.

The **availability of data** or fragmentation of data is one of the biggest problems healthcare organizations face. Ensuring stakeholders', providers', and patients' data access is critical. Patients "own" their data, and as a result, access needs to be transparent, quick, and efficient for all. Data access governance may set permissions and authentication levels for user roles to systems.

Finally, the data and information governance for a project may state the types of data content collected. This may include health data, metadata, and location, profile, and behavior data.

Security

There are significant risks in the security and privacy of patient data. With sensitive health data, there is rightly an increased sensitivity to security and privacy concerns. The tools used for analysis, storage, management, and utilization of data are from a variety of sources, which exposes the data to risk. Significant data breaches have occurred in the healthcare industry by reputable organizations. In the UK, the Information Commissioner (ICO) fined the Brighton and Sussex University Hospitals NHS Trust after it was found that the sensitive data of several thousand people were resident on hard drives placed up for auction on eBay (Lambert 2018). American health insurance company Anthem was responsible for the most substantial healthcare breach to date, exposing the personal records, including home addresses and personal information of over 70 million current and former members (Rabin 2016).

Data loss, including data theft, is a genuine problem that is continuously tested to its limits. Security challenges include authentication for every team and member accessing datasets, restricting access based on user needs, recording data access histories, and meeting compliance regulations, and ensuring data is encrypted to protect from snoopers and any malintent. The security of medical devices poses a unique threat because of their technological diversity. Medical devices, from health applications on a smartphone to insulin pumps, are becoming increasingly networked, leaving unique openings for hackers (Independent 2019a). This kind of activity is regarded as dangerous, although for those more technical, it is understood that this can be utilized for positive benefit. Nevertheless, the concern remains that if exploited, similar device vulnerabilities could lead not only to data breaches but fatalities in people relying on medical devices. Vendor devices, services, and APIs are only as good as the developers that have developed them, making the balance between disruption and progress one of real-world experience.

Ethics of AI/Data Science

Data science ethics comprises three strands: ethics of data, ethics of intelligence, and ethics of practices (Mittelstadt et al. 2016). Novel solutions to problems developed through machine learning are themselves leading to questions of morality and ethics, and governance is moving at the pace of the industry itself. A primary application of machine learning in healthcare involves patient diagnosis and treatment. AI models are deployed to help physicians diagnose patients, especially in cases involving relatively rare diseases or when outcomes are hard to predict. There are many scenarios within AI for which there are no precedents, regulations, or laws. As such, it is paramount for developers and practitioners alike to consider the ethical and moral implications of creating intelligent systems.

One of the biggest pitfalls of AI is bias, and there are several forms of bias.

Machine learning bias, also known as algorithm bias or AI bias, is a phenomenon that occurs when an algorithm produces results that are systematically

prejudiced due to erroneous assumptions in the machine learning process. Machine bias can be the result of several causes, such as the creator's bias on data used to train the model. Biases are often spotted as first-time problems with subtle and apparent ramifications. All machine learning algorithms rely on a statistical bias to make predictions about unseen data. However, machine bias reflects a prejudice from the developers of the data. The capabilities of AI regarding speed and capacity of processing far exceed that of humans. Therefore, it cannot always be trusted to be fair and neutral. Google are leaders in AI, as seen with Google Photos service, where AI is used to identify people, objects, and scenes. However, it can still go wrong, such as when the search engine showed insensitive results for related searches for white and black teenagers. Software used to predict future criminals have been demonstrated to show bias toward black people (The Guardian 2016).

Data bias occurs when the available data is not representative of the study population or phenomenon. It may also denote that the data does not include features that capture the phenomenon we want to predict.

It is noteworthy that artificially intelligent systems are created by humans – who are biased and judgmental. It is plausible that for as long as humans are involved in decisions, bias will always exist. This is **human bias**. Microsoft launched an AI chatbot named Tay in 2017. Tay interacted with other Twitter users and learned from its interactions with others. Once the Twittersphere seized hold of Tay, trolls steered the conversation from positive interactions to interactions with trolls and comedians. Within hours, Tay was tweeting sexist, racist, and suggestive posts meaning Tay only lived for 24 h. This experiment highlights the question of whether AI can ever really be safe if it learns from human behavior (Telegragh 2017).

Human thinking AI systems have demonstrated bias amplification and catalyzed conversations surrounding the ethical use of AI technology. Early thinking systems built on population data showed significant signs of bias regarding sex, race, social standing, and other issues. An infamous example of algorithmic bias can be found in criminal justice systems. The Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) algorithm was dissected in a court case about its use in Wisconsin. Disproportionate data on crimes committed by African Americans were fed into a crime prediction model, which then subsequently output bias toward people from the black community. There are many examples and definitions of biased algorithms (Dressel and Farid 2018). Algorithms that assess home insurance risk, for instance, are biased against people who live in particular areas based on claim data.

Data normalization is a crucial factor in mitigating these concerns. If data is not normalized for such sensitivities and systems not correctly validated, humanity runs the risk of underrepresenting and skewing machine learning models for minorities and underrepresenting many groups of people. Removing bias does not mean that the model will not be biased. Even if an absolute, unbiased model were to be created, there is no guarantee that the AI will not learn a specific or particular bias.

Correcting bias begins with the acknowledgment that bias exists. The data science community must actively work to eliminate bias. Engineering must honestly question preconceptions toward processes, intelligent systems, or how bias may expose itself in data or predictions.

Data ethicist James Moor defined implicit and explicit ethical agents (Moor 2009). Implicit agents are deemed ethical because of their basic programming or purpose, whereas explicit agents are machines given principles or examples to learn from to make ethical decisions in uncertain or unknown circumstances. Overcoming bias can involve post-processing regarding model calibration. Data resampling can help smoothen a skewed sample. For many reasons collecting more data is not very easy and can cause resource problems.

Bias raises a philosophical question on the premise that machine learning assumes that biases are generally harmful. However, imagine a system that is interpreted by its evaluators to be biased, and so the model is retrained with new data. If the model were to output similarly biased results, the evaluation might wish to consider that this is an accurate reflection of the output and hence require a reconsideration of what bias is present. This is the beginning of a societal and philosophical conflict between two species.

Taking a cue from the software quality assurance world which assumes that bugs exist in software and need to be fixed, those working on AI models should assume their data has built-in or systemic bias and other issues, and that formal processes are needed to detect, correct, and address. Detecting bias and ensuring fairness doesn't come easy and is most effective when subject to review and validation from a diverse set of perspectives.

That means building intentional diversity, cognitive diversity, socioeconomic diversity, cultural diversity, and physical diversity into the processes used to detect unfairness and bias to help improve the process and mitigate the risk of missing something critical.

Diversity in the workplace can prevent bias from creeping into intelligence. Should the researchers and developers creating our AI systems themselves lack diversity, and then the problems that AI systems solve and training data used both become biased based on what is fed into AI training data. Diversity ensures a spectrum of thinking, ethics, and mind-sets and promotes AI models that are less biased. Although algorithms may be written to best avoid biases, doing so is extraordinarily challenging. For instance, even the motives of the people programming, AI systems may not match up with those of physicians and other caregivers who can invite bias.

First-Time Problems/Unintended Consequences

As the adoption of AI in medicine becomes commonplace, first-time problems and unintended consequences will govern the media perception and direction of AI ethics. There are very few technologies that have been embedded in healthcare without unintended or adverse effects. The question quickly comes to how data scientists, on behalf of humanity, can mitigate these risks.

The ethics of AI and unintended consequences have been significantly directed by *The Three Laws*, published by science fiction writer Isaac Asimov in 1942 (Asimov 1950). Asimov's Laws, found in the "*Handbook of Robotics*, 56th Edition, 2058 A.D.," were designed as a safety feature to prevent robots from harming humans:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given by a human being unless it conflicts with the First Law.
3. A robot must protect its existence as long as such protection won't conflict with the First or Second Law.

In Asimov's stories, manlike machines behave in counterintuitive ways. These acts are the unintended consequence of how the agent applies Asimov's Laws to its environment. Only 70 years later, Asimov's science fiction fantasy is turning into reality. In 2012, the IEEE and British Standards Institute both published best practice guides for ethical agent engineering (Bryson and Winfield 2017).

Best practice guides to designing ethically sound agents are typically grounded in Asimov's Laws. Fundamentally, intelligent agents are made of binary code and do not contain or obey Asimov's Laws. Their human architects are responsible for implementing the laws and reducing the risk of unintended consequences. AI is unlikely to become evil and turn on humanity in the way depicted by Hollywood. Instead, a lack of context could lead to AI making unintended and disastrous actions. For instance, an intelligent agent that was tasked with eradicating HIV in the global population could eventually come to some conclusion that to achieve its objective, and it should kill everyone on the planet. It is easier to frame AI intelligence in a harmful setting. Without careful management, an AI agent's utility function could allow for potentially harmful scenarios. There is a limited foundation to suppose that an AI agent would have such an extreme adaptation, yet it is worth consideration. There are undoubtedly many positive unintended consequences to AI that could save humanity from itself.

First-time problems are a form of unintended consequence. Data can now be used within AI to detect risks of conditions such as depression, anxiety, hypertension, and pancreatic cancer. If an application was able to tell one about a terminal disease, one could do nothing about, would one want to know? Would one even want to know that there was a choice to know? For instance, if someone were to be informed they had a reduced risk of developing ill-health, there may be some who would then have a propensity for riskier behaviors. If a patient is told their susceptibility to lung cancer is lower than average, would this lead to a higher risk of them becoming smokers or other risky behaviors? Similarly, knowing the risk of developing particular conditions could lead to mental and emotional consequences. The implications on human psychology and behavior of knowing disease risk will be discovered as time and evidence develops.

Preventing Immoral Algorithms

Machine deception is a serious challenge. Algorithms can, and do, act immorally. Deepfakes have been successfully used to manipulate in pornography, politics, acting, and social media. Use of deepfakes to spoof others is a potential minefield for healthcare. (Deepfakes refer to images or videos which are replaced with someone else's likeness using artificial neural networks.) Blockchain technology to verify content from trusted sources may prove to be a partial fix. However as deepfakes improve, trust in digital content diminishes.

AI algorithms have performed in unethical ways typically due to design. Uber's Greyball algorithm attempted to predict which passengers were undercover police officers and used it to identify and deny transport (Isaac 2017). Volkswagen's algorithm allowed vehicles to pass emission tests by reducing nitrogen oxide emissions during the test phase (Contag et al. 2017). Both organizations were internationally condemned for their public deception and lack of transparency. Precedents such as these from the world's largest companies demonstrate that internal and external auditing is required to validate the integrity and ethics of algorithms and, indeed, their organizations.

There is a genuine concern that AI may learn to act immorally not only from its creators but also from its experience. The Ecole Polytechnique Fédérale of Lausanne's Laboratory of Intelligent Systems in Switzerland conducted a project that monitored robots designed to search for positive resources and ignore dangerous items collaboratively (Laboratory of Intelligent Systems 2019). Robots were designed as genetic agents equipped with sensors and a light that was used to flag the identification of a positive resource, which was finite in number. Each agent's genome dictated its response to stimulus and was subject to hundreds of generations of mutations. Agent learning was reinforced by the agent receiving positive marks for identifying positive resources and negative marks for its proximity to poisonous items. The top 200 highest performing genomes were mated and mutated at random to produce the next generation of agents. In their first generation, agents switched their lights on when they discovered a positive resource. This enabled other agents to find the positive resource. Due to the limited number of positive resources, not all agents were able to benefit. Overcrowding meant that some agents would also be distanced from the positive resource they initially found. By the 500th generation, the majority of agents had evolved to keep their light switched off when they discovered a positive resource, and a third of agents evolved to act in a way that was the exact opposite of their programming. Some agents had grown to identify lying agents that had an aversion to the light. Agents that were initially designed to cooperate eventually ended up lying to each other due to scarcity. These concerns are amplified when applied to incentivized clinical decision-support systems that could generate increased profits for their architects or providers – a system that was recommending clinical tests, treatment, or devices in which they hold a stake, or by altering referral patterns. In healthcare, this scenario is very concerning. All machine learning models, regardless of their use, must be governed and verifiable.

A machine learning model that was incentivized to make decisions should also adhere to robust standards of transparency, morality, and scrutiny.

Informed Consent

Informed consent refers to the user (or patient) being aware of what their data will be used for. Informed consent refers to an individual being legally able to give consent. Typically this requires an individual to be over 18, of sound mind, and able to exercise choice. Consent should ideally be voluntary.

Freedom of Choice

Freedom of choice refers to the autonomy to decide whether data is shared and with whom. A person's choice as to whether to share their data could result in a future where people are exempt from opportunities until otherwise demonstrated by data. In an ideal world, each person should have a choice as to whether to share their data. In practice, this is neither realistic nor plausible.

The advances of AI and machine learning in healthcare mean that today there are two groups of people: those that seek out health information to help them plan and manage future scenarios and those that are happy to live in a state of not knowing. As the datafication of everything continues, those that are happy to live in a state of ignorant bliss are finding less opportunity to do so.

Data Ownership

Data ownership is an example of a complex first-time problem that has facilitated the development of international policy and governance. Electronic health records enable machine learning-based predictive analytics that enables providers to provide enhanced levels of care. Although it seems unlikely a patient would not want to not share their health data for improved morbidity and mortality, one would reasonably assume it would become more difficult to withdraw such privileges, much akin to closed-circuit television (CCTV) where consent is not often noted or obvious and is widely assumed to have utilitarian benefits over the long term. Many platforms prevent users from accessing their full range of services if data sharing is disabled. This is driven by the organizational need to store data in a robust and central repository for safety, governance, and improvement. Data collection, usage, and sharing have become a fundamental part of supporting data-driven quality improvement processes. Historically, user data has been owned by companies rather than the individual, with some organizations threatening to sue individuals taking copies of their device-generated data (BoingBoing 2019).

How Data Affects the Future

The sharing of patient data and aggregation of big datasets is being used to enhance diagnosis, treatment, and care. As the types and quality of data improve, healthcare's precision will become more precise in the areas discussed next. Prioritizing Treatments Big data medical sets can enable predictive analytics to determine optimal treatment pathways for various segments of the population. There is the potential to prioritize only those who are likely to see efficacy in the treatment, which begs the question as to how best to support those that do not.

The analysis of real-world experience data, clinical studies, randomized clinical trial (RCTs), and pharmacological data is facilitating treatment and management discovery. Digital health interventions have been demonstrated to place type 2 diabetes into remission (reversal) and reduce the number of seizures suffered by patients with epilepsy. Data has the potential to develop new pharmacological interventions and disrupt traditional treatment paradigms. The use of real-world evidence (RWE) at scale from patient communities, digital education programs, and health tracking apps is demonstrated to improve patient self-management of care (Summers and Curtis 2020).

A concern with real-world evidence has typically been a lack of academic robustness. However, RWE is increasingly used in research to determine population usage and benefits. RWE is a vital part of AI's ethical journey, and there must be trust between patient and provider. Previous research has found that self-reported health outcomes can be quite close to actual values (McAdams et al. 2007; Løvaas et al. 2015).

The precise audiences required for RCTs and academic studies can be identified quicker and more accessible through digital platforms. This has benefits, such as quicker project recruitment times, more significant potential, and multisource comparison. Real-world data is also being used to develop better and more effective drugs.

How Machines Affect Our Behavior

AI's rapid evolution is prompting humankind to evaluate just what it means to be a human. Virtual assistants such as Alexa, Siri, and Google Home are in the hands of over 100 million people – and voice-based telephony agents are commonplace. Not only are agents able to mimic conversational language, but emotion and reaction modeling is more sophisticated and believable than ever. In 2015, a bot named Eugene won the Turing Challenge, which asks humans to rate the quality of their communication with an unknown agent and to guess whether it is a human or an AI agent. Eugene was able to disguise itself as a human to more than half of its human raters (University of Reading 2018). The blurring between humankind and robot-kind is imploring exploratory interspecies relationships. In 2017, Chinese AI engineer Zheng Jiajia married robot Yingying, a robot spouse he created to ease the pressure of marriage (Huffington Post 2018).

Human-replacement sex robots are now conventional, with connected AI robots developed with humanlike skin and warmth, as well as safety measures to mitigate electrical shocks or worse. Whether applied to banking, healthcare, transport, or anything else, humanity is at the beginning of a juncture where humans engage with AI agents as frequently as though they too are humans. Where humans have limits as to the amount of attention, kindness, compassion, and energy they can expend, AI bots are an infinite resource for developing and maintaining relationships.

Behaviors are already influenced and manipulated by technology. Web landing pages shared links, and user experiences are optimized through multivariant testing. This is a crude algorithmic approach to capture human attention. In the 2017 US Elections, Donald Trump's campaign famously used and tested more than 50,000 ad variations daily to micro-target voters (The Guardian 2018). Human experimentation has been taking place for centuries, so this is nothing new. However, there is now the ability to influence choice and freedom like never before. Organizations have a moral responsibility to analyze the impact of AI and safeguard particular groups of people, including vulnerable adults and children, from manipulative tools like nudging or social proofing. Codes of conduct for behavioral experimentation within technological experiments are required. Technology addiction is becoming a growing concern. Robert Lustig, a professor at the University of Southern California, discovered that the human brain responds to technology in a similar way to other addictive substances (Lustig 2013; Cargill 2016). Internet addiction has been documented in a variety of countries, and adolescents appear to be more at risk, partly due to the prefrontal cortex being the last part of the brain to develop. According to the NHS, one in three people is addicted to something (NHS 2019). In 2015, a 38-year-old man was found dead after playing video games for 5 days in an Internet cafe in Taipei (The Sydney Morning Herald 2016). Real-world and academic evidence demonstrates that technology addiction is a mounting issue. AI has the potential to improve productivity and discovery. However, humankind is responsible for taking measures to prevent AI leading to destructive, digital addictions that make us more insular, overdependent, and lethargic.

Conclusions

Wearables and the health IoT hold the key to a truly connected healthcare with data captured by sensors playing an increasingly dominant role. Many factors are accelerating the acceptance of wearable healthcare solutions, particularly their use in clinical trials and academic studies to monitor patient health and lifestyle factors. Participants use apps to record their lifestyle habits, nutrition, activity, and medication adherence and to monitor medication side effects, among others. Wearables and real-world data are beginning to be used by insurance companies to incentivize wellness. Insurance companies have historically targeted such insurance products toward digital-savvy buyers, offering the latest gadgets to incentivize health improvements. However, incentivized insurance products are proving particularly useful in combat against noncommunicable diseases. The delivery of connected,

digital healthcare opens an opportunity to monitor, manage, and reverse certain noncommunicable diseases, broadening the risk portfolio and creating a longer-living, reduced morbidity population.

As sensors become faster, smaller, and more capable, patient health records profiles will eventually comprise detailed sleep analyses, details of continuous blood glucose monitoring, heart rate, blood pressure, and approximate calorie burn. A smartwatch will combine a variety of diagnostic tools, able to monitor blood pressure, heart rate variability, blood glucose, ketones, and more. Health sensors will become embeddable, biodegradable, and continuously connected, playing critical roles in such tasks as patient care.

While AI and digital health hold significant promise, healthcare is facing significant challenges in coping with the rapid drive to integrate these systems into medicine – where the stakes are high. Meaningful changes in healthcare will come from the right blend of innovation and deliberation.

As the boundaries of discovery are tested, the potential of providing patients precise evidence-based medicine and care is being realized. At the same time, there is an obligation to explore and anticipate the social, economic, political, and ethical consequences of innovation. Big data and AI's adoption in healthcare is best managed as a gradual development, which would be useful for stakeholders to have time to map and understand the possible consequences of its implementation. Robust standards are required for auditing and understanding AI models for upholding quality assurance and trust.

Collaborative academic research, clinical evidence, and policymaking will shape the domain of AI and healthcare. The development of AI standards requires a diverse and public forum that is academically robust and reviewed periodically in line with the latest evidence base. Awareness within the medical professional and patient communities is required. Healthcare professionals need basic knowledge of how AI applications function in their respective medical settings to understand the advantages, disadvantages, and consequences of how AI can facilitate and assist them in their everyday tasks. Patients must become accustomed to engaging with AI systems and explore the benefits for themselves. Demonstratable examples of how AI in healthcare is optimizing efficiency, optimizing resources, and improving population health outcomes will strengthen the public perception and adoption of AI.

Organizations working with AI need to communicate and engage the general public about the potential advantages and risks of using AI in medicine. Organizations developing AI must work with academia to take the necessary steps to be able to measure the success and the effectiveness of AI systems independently. It is vitally important to encourage affordable AI solutions to democratize AI as the stethoscope of the twenty-first century and published validated evidence and literature are essential to AI's trust and adoption.

With more data now than ever before, the most promising era for healthcare is yet to come.

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